1.50 Bivariate Data: PMCC and Linear Regression

Worked Solutions

1)

Strong Positive Correlation

2)

Moderate Negative Correlation

3)

$r = 0.864$

4)

$r = -0.391$

5)

The product moment correlation coefficient, r, is such that $-1 \leq r \leq 1$, so r cannot be equal to $-\frac{7}{3}$ or $\frac{8}{7}$.

6)

Explanatory (independent) variable = Hours of exercise per week Response (dependent) variable = Levels of stress

7)

Probably correct. You would expect that as age increases, reaction time would also increase.

8)

Definitely incorrect. The PMCC, r, must be $-1 \leq r \leq 1$, so you cannot have a PMCC of 1.13.

9)

Definitely correct. The line going through (13, 7) and (20, 11) would have a positive gradient, and seeing as there are only two points the PMCC would be precisely 1.

10)

$r = 0.995$ (3 SF)

11 a)

$r = 0.934$ (3 SF)

11 b)

There is a strong positive correlation between the number of hours the students studied in a week and the students' test scores.

12)

A PMCC of -0.76 shows a moderately strong negative correlation between the frequency of smoking and respiratory function. So as the amount of smoking increases, the respiratory function decreases.

13 a)

Winchester appears to be an outlier. It may have an unusually high average house price due to its high quality of life / historical significance / transport links to London, etc.

13 b)

Correlation is how close the data is to a straight line. So the data does not appear to be well-correlated, but there appears to be a negative association (non-linear as the data is curved).

13 c)

As the distance from London increases, the average house price decreases.

14 a)

Data point A is unlikely to be correct as the runner did zero exercise the week before but has still managed to run a long distance. It may well be the case, however, that the runner has considerable prior experience of running long distance and does not require as much exercise in the lead-up.

14 b)

Data point B could represent a new member of the running group. As they are already on week 10, it may well be that B has only just joined the group and hence needs to put in more exercise the week before to reach a similar standard in the run as others in the group.

15 a)

Data point A has a zero score for the Spelling test. This may be accurate as the student may have been absent.

15 b)

Data point B has a much higher Arithmetic test score than any of the other students. All of the other students scored under 100, so it may well be that the tests had a maximum score of 100. It may well have been inputted incorrectly when the graph was drawn.

16)

The claim is incorrect as the negative correlation in the scatter diagram does not imply that amount of sleep and the task completion time are causally linked (correlation does not imply causation).

17)

The data in the diagram is close to a straight line, and the PMCC of -0.979 is very strong (negative) correlation.

18)

$y = 2.88x + 2.84$

19)

$y = 773 - 3.90x$

20)

For every £1 spent each month in advertising, the sales revenue increases by £1.96.

21)

For every 1 penny increase in price, the number of cans sold is reduced by 2.22 cans.

22)

The fastest time will be recorded by the person that sleeps the longest. If a runner sleeps for 82 hours, they will finish the 100 m race instantaneously.

23 a)

Marks scored in the final exam per hour of lunchtime workshop attended.

23 b)

This will increase the average student's final score on the exam by $1.12 \times 2.5 = 2.8$.

23 c)

$1 \times 5 \times 38 = 190$ So there is a maximum number of 190 hours of lunchtime workshop. So $t = 200$ would be beyond the number of hours available, and it would also give $98 + 1.12 \times 200 = 322$ which is a higher score than the maximum possible score for the exam (which was 300).

24 a)

3 days is 6×12 hours, so $t = 6$. $0.67 \times 6 = 4.02$, so on average the fungus will have grown by 4.02 mm.

24 b)

5 cm = 50 mm. $50 = 3.1 + 0.67t \Rightarrow t = 70$ So 35 days

25 a)

$y_A = 0.0497 \times 132 - 0.0277 = 6.5327$ tonnes per hectare

25 b)

$y_B = 0.0497 \times 121 - 0.0277 = 5.986$ tonnes per hectare

25 c)

The estimate calculated for Farm A uses interpolation as 132 is within the data collected, and so is more reliable than the estimate calculated for Farm B as that uses extrapolation as 121 is outside of the data collected.